QUESTION 1

```cpp
#include <iostream>
#include <iomanip>
#include <string>

using namespace std;

// Struct to hold gadget information
struct Gadget {
	string name;
	double price;
};

// Gadget catalog
Gadget catalog[] = {
	{"Wireless Mouse", 150.00},
	{"Bluetooth Speaker", 250.00},
	{"USB Flash Drive", 100.00},
	{"Gaming Keyboard", 500.00},
	{"Noise-Cancelling Headphones", 750.00},
	{"Smartwatch", 1200.00},
	{"External Hard Drive", 1000.00},
	{"Portable Charger", 300.00},
};

const int CATALOG_SIZE = 8;
const double DISCOUNT_THRESHOLD = 2000.00;
const double DISCOUNT_PERCENTAGE = 0.15;

int main() {
	string firstName, lastName;
	int itemCount, itemNumber, quantity;
	double subtotal = 0.0, discount = 0.0, finalTotal = 0.0;

	// Prompt user for name
	cout << "Enter your first name: ";
	cin >> firstName;
	cout << "Enter your last name: ";
	cin >> lastName;
	cout << endl;

	// Display catalog
	cout << "Tech Store Catalogue: \n";
	for (int i = 0; i < CATALOG_SIZE; i++) {
		cout << i + 1 << ". " << catalog[i].name << " - R"
			<< fixed << setprecision(2) << catalog[i].price << endl;
	}
	cout << endl;

	// Get number of items to purchase
	cout << "How many items would you like to purchase (up to 10)? ";
	cin >> itemCount;
```

```
	// Validate number of items
	if (itemCount < 1 || itemCount > 10)
	{
		cout << "Invalid number of items. Exiting program.\n";
		return 1;
	}

	// Process item purchases
	for (int i = 0; i < itemCount; i++)
	{
		cout << "Enter the item number (1-8): ";
		cin >> itemNumber;

		// Validate item Number
		if (itemNumber < 1 || itemNumber > CATALOG_SIZE) {
			cout << "Invalid item number. Please try again.\n";
			i--; // Repeat this iteration
			continue;
		}

		cout << "Enter the quantity: ";
		cin >> quantity;

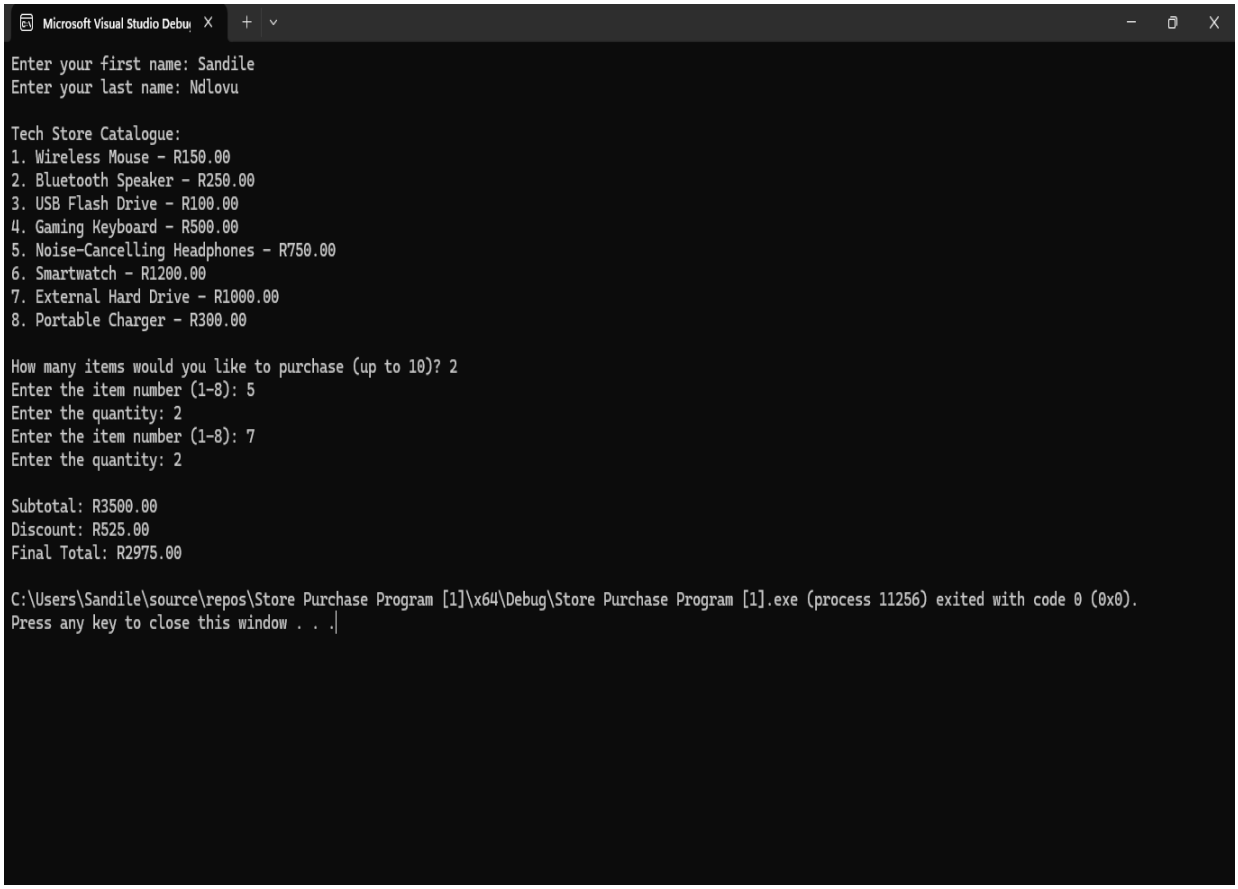
		// Validate quantity
		if (quantity < 1) {
			cout << "Invalid quantity. Please try again.\n";
			i--; // Repeat this iteration
			continue;
		}

		// Calculate subtotal
		subtotal += catalog[itemNumber - 1].price * quantity;
	}

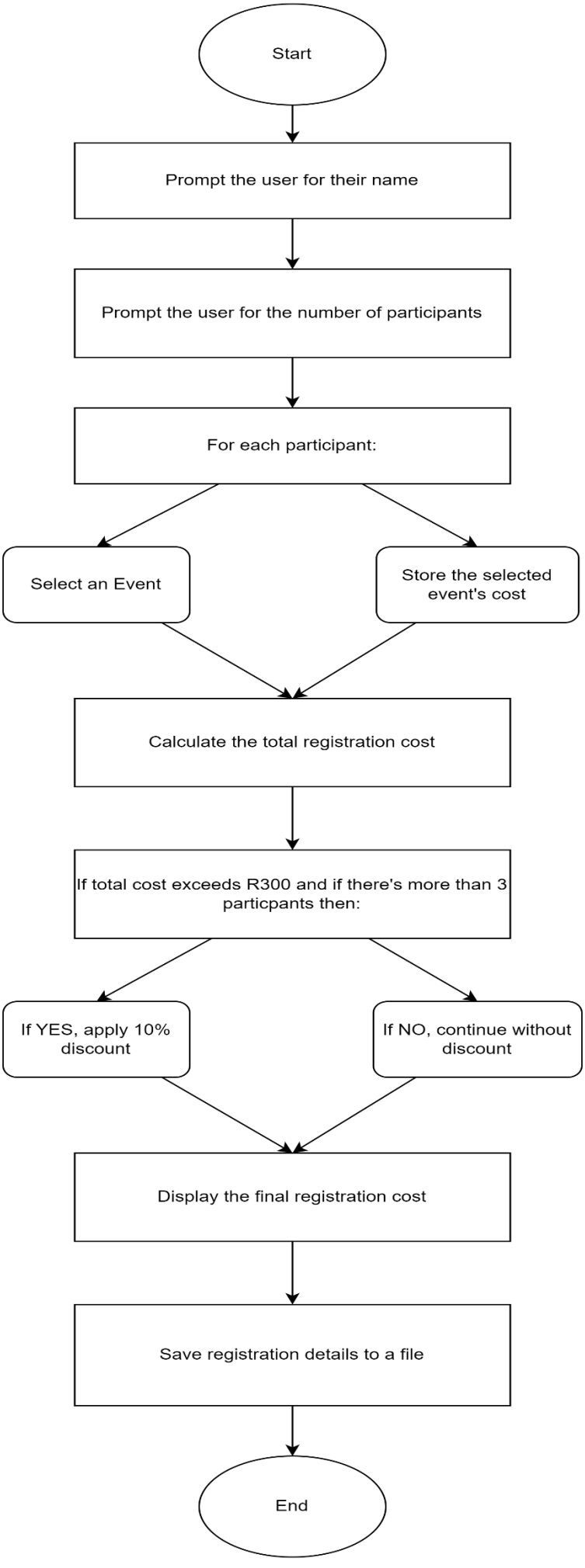
	// Apply discount if applicable
	if (subtotal > DISCOUNT_THRESHOLD) {
		discount = subtotal * DISCOUNT_PERCENTAGE;
	}
	finalTotal = subtotal - discount;

	// Display results
	cout << fixed << setprecision(2);
	cout << "\nSubtotal: R" << subtotal << endl;
	cout << "Discount: R" << discount << endl;
	cout << "Final Total: R" << finalTotal << endl;

	return 0;
}
```


Microsoft Visual Studio Debu
Enter your first name: Sandile
Enter your last name: Ndlovu
Tech Store Catalogue:
1. Wireless Mouse - R150.00
2. Bluetooth Speaker - R250.00
3. USB Flash Drive - R100.00
4. Gaming Keyboard - R500.00
5. Noise-Cancelling Headphones - R750.00
6. Smartwatch - R1200.00
7. External Hard Drive - R1000.00
8. Portable Charger - R300.00
How many items would you like to purchase (up to 10)? 2
Enter the item number (1-8): 5
Enter the quantity: 2
Enter the item number (1-8): 7
Enter the quantity: 2
Subtotal: R3500.00
Discount: R525.00
Final Total: R2975.00
C:\Users\Sandile\source\repos\Store Purchase Program [1]\x64\Debug\Store Purchase Program [1].exe (process 11256) exited with code 0 (0x0).
Press any key to close this window . . .

QUESTION 2


Start
Prompt the user for their name
Prompt the user for the number of participants
For each participant:
Select an Event
Store the selected event's cost
Calculate the total registration cost
If total cost exceeds R300 and if there's more than 3 particpants then:
If YES, apply 10% discount
If NO, continue without discount
Display the final registration cost
Save registration details to a file
End

2.2) **ALGORITHM FOR SPORTS EVENT REGISTRATION SYSTEM:**

1. Start
2. Prompt the user to enter their name
3. Prompt the user to enter the number of participants
4. Initialize total_cost to 0
5. For each participant:
    - Display the list of events and their prices
    - Prompt the user to select an event
    - Add the selected event's cost to total_cost
6. If total_cost > 300 AND number of participants >= 3, apply a 10% discount:
    - total_cost = total_cost * 0.9
7. Display the final total cost
8. Save the registration details (name, number of participants, event choices, final cost) to a file
9. End

QUESTION 3

```cpp
#include <iostream>
#include <iomanip> // For setting precision
using namespace std;

int main() {
	double temperatures[7]; // Declare an array to store temperatures
	double sum = 0; // Variable to store sum of temperatures
	double highest, lowest; // Variables to store highest and lowest
temperatures

	// Input temperatures for 7 days
	cout << "Enter the temperature for 7 days:" << endl;
	for (int i = 0; i < 7; i++) {
		cout << "Day " << i + 1 << ": ";
		cin >> temperatures[i];
		sum += temperatures[i];

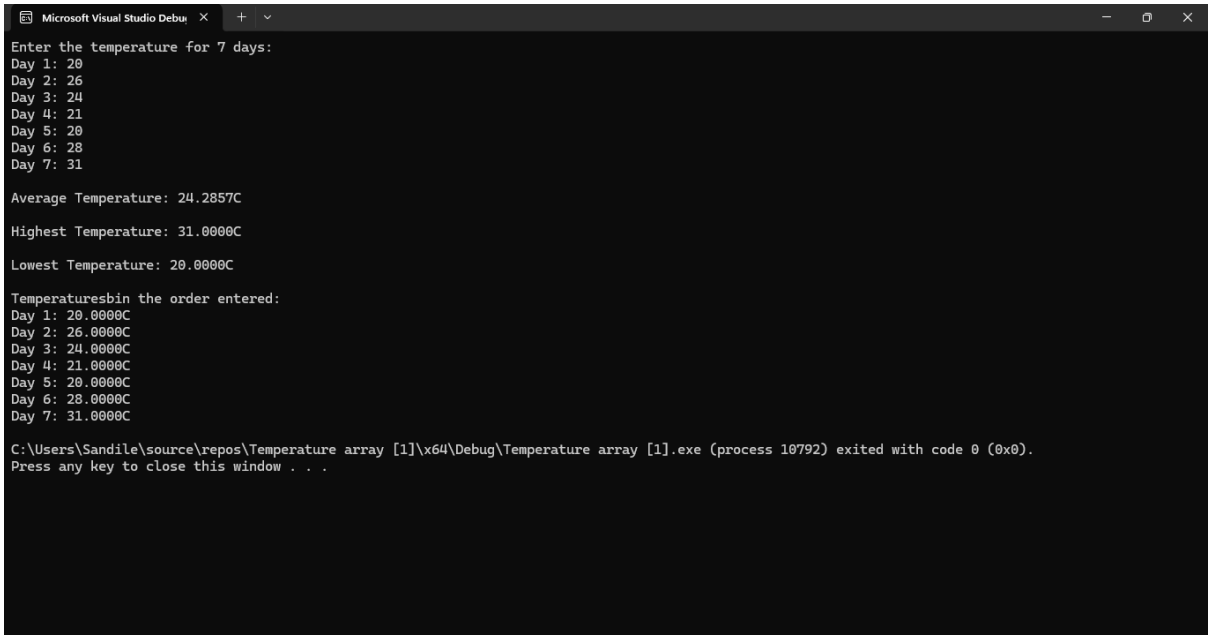
		// Assign first temperature as highest and lowest initially
		if (i == 0) {
			highest = lowest = temperatures[i];
		} else {
			if (temperatures[i] > highest) highest = temperatures[i];
			if (temperatures[i] < lowest) lowest = temperatures[i];
		}
	}

	// Calculate the average temperature
	double average = sum / 7;

	// Display results
	cout << fixed << setprecision(4); // Set precision to 4 decimal places
	cout << "\nAverage Temperature: " << average << "C" << endl;
	cout << "\nHighest Temperature: " << highest << "C" << endl;
	cout << "\nLowest Temperature: " << lowest << "C" << endl;

	// Display temperatures in the order entered
	cout << "\nTemperaturesbin the order entered:" << endl;
	for (int i = 0; i < 7; i++) {
		cout << "Day " << i + 1 << ": " << temperatures[i] << "C" << endl;
	}

	return 0;
}
```


Microsoft Visual Studio Debu
Enter the temperature for 7 days:
Day 1: 20
Day 2: 26
Day 3: 24
Day 4: 21
Day 5: 20
Day 6: 28
Day 7: 31
Average Temperature: 24.2857C
Highest Temperature: 31.0000C
Lowest Temperature: 20.0000C
Temperaturesbin the order entered:
Day 1: 20.0000C
Day 2: 26.0000C
Day 3: 24.0000C
Day 4: 21.0000C
Day 5: 20.0000C
Day 6: 28.0000C
Day 7: 31.0000C
C:\Users\Sandile\source\repos\Temperature array [1]\x64\Debug\Temperature array [1].exe (process 10792) exited with code 0 (0x0).
Press any key to close this window . . .

QUESTION 4

```cpp
#include <iostream>
#include <string>

using namespace std;

struct Member {
	string name;
	string membershipNumber;
	int sessionCredits;
};

void createAccount(Member& member) {
	cout << " --- Create Membership Account ---" << endl;
	cout << "Enter your name: ";
	getline(cin, member.name);
	cout << "Enter your Membership number: ";
	getline(cin, member.membershipNumber);

	do {
		cout << "Enter initial session credits (must be at least 1): ";
		cin >> member.sessionCredits;
		if (member.sessionCredits < 1) {
			cout << "Invalid input! Initial credits must be at least 1. "
<< endl;
		}
	} while (member.sessionCredits < 1);

	cout << "Membership account created successfully!" << endl;
}

void addSessionCredits(Member& member) {
	int creditsToAdd;
	cout << "--- Add Session Credits ---" << endl;
	do {
		cout << "Enter the number of session credits to add (must be
positive): ";
		cin >> creditsToAdd;
		if (creditsToAdd <= 0) {
			cout << "Invalid input! Session credits must be positive." <<
endl;
		}
	} while (creditsToAdd <= 0);

	member.sessionCredits += creditsToAdd;
	cout << "Session credits added successfully! Current credits: " <<
member.sessionCredits << endl;
}

void bookSession(Member& member) {
	cout << "--- Book a Training Session ---" << endl;
	if (member.sessionCredits > 0) {
		member.sessionCredits--;
		cout << "Training session booked successfully! Remaining Credits: "
<< member.sessionCredits << endl;
	} else {
		cout << "Insufficient session credits! Please add more credits to
book a session. " << endl;
	}
}

void checkCredits(const Member& member) {
```

```
	cout << "--- Check Session Credits ---" << endl;
	cout << "Your current session credits are: " << member.sessionCredits <<
endl;
}

void displayAccountDetails(const Member& member) {
	cout << "--- Display Account Details ---" << endl;
	cout << "Member Name: " << member.name << endl;
	cout << "Membership Number: " << member.membershipNumber << endl;
	cout << "Session Credits: " << member.sessionCredits << endl;
}

int main() {
	Member member;
	bool accountCreated = false;
	int choice;

	do {
		cout << "\n--- Fitness Membership Management System ---" << endl;
		cout << "1. Create Membership Account" << endl;
		cout << "2. Add Session Credits" << endl;
		cout << "3. Book a Training Session" << endl;
		cout << "4. Check Session Credits" << endl;
		cout << "5. Display Account Details" << endl;
		cout << "6. Exit" << endl;
		cout << "\nChoose an option: ";
		cin >> choice;
		cin.ignore(); // To handle newline characters

		switch (choice) {
		case 1:
			createAccount(member);
			accountCreated = true;
			break;
		case 2:
			if (accountCreated) {

				addSessionCredits(member);
			} else {
				cout << "Please create an account first!" << endl;
			}
			break;
		case 3:
			if (accountCreated) {

				bookSession(member);
			} else {
				cout << "Please create an account first" << endl;
			}
			break;
		case 4:
			if (accountCreated) {

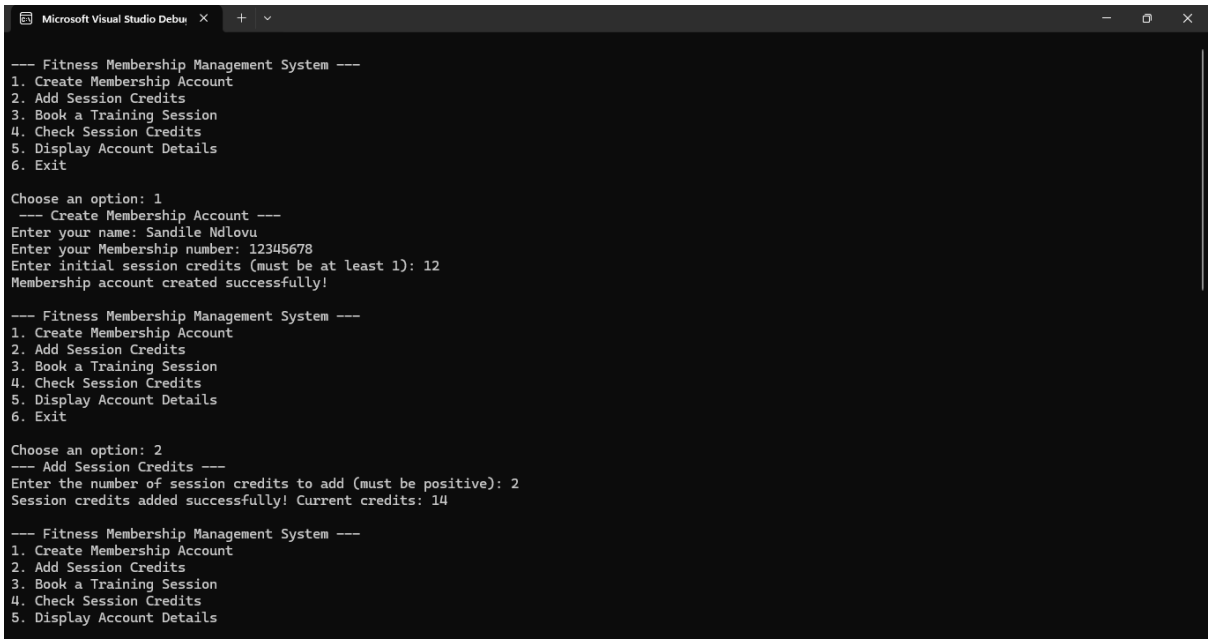
				checkCredits(member);
			} else {
				cout << "Please create an account first" << endl;
			}
			break;
		case 5:
			if (accountCreated) {

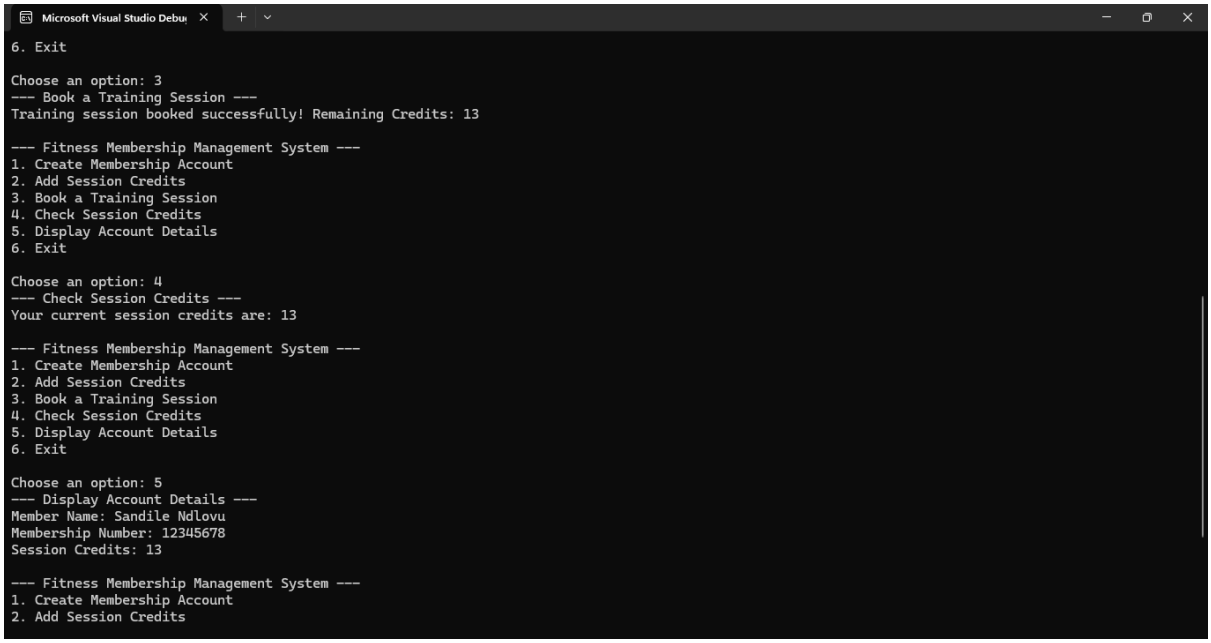
				displayAccountDetails(member);
			} else {
```

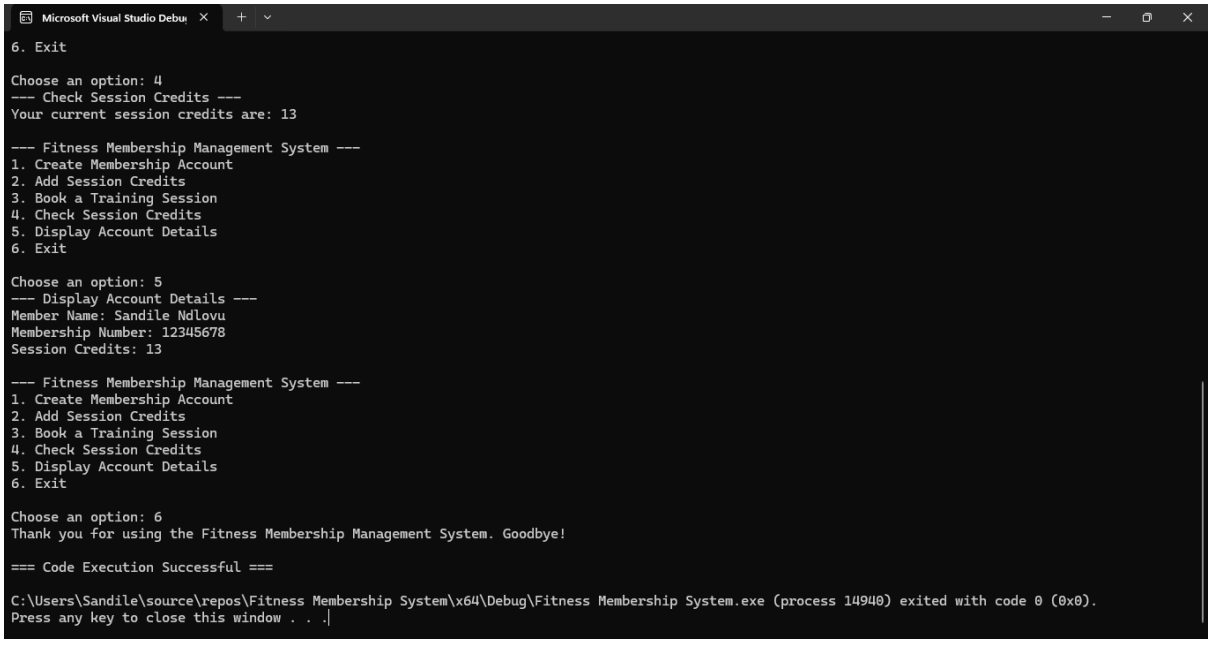
```
                        cout << "Please create an account first" << endl;
                    }
                    break;
            case 6:
                    cout << "Thank you for using the Fitness Membership
Management System. Goodbye!" << endl;
                    break;
            default:
                    cout << "Invalid option! Please choose a valid option." <<
endl;
            }
      } while (choice != 6);

      cout << "\n=== Code Execution Successful ===" << endl;

      return 0;
}
```




Microsoft Visual Studio Debu
6. Exit
Choose an option: 3
--- Book a Training Session ---
Training session booked successfully! Remaining Credits: 13
--- Fitness Membership Management System ---
1. Create Membership Account
2. Add Session Credits
3. Book a Training Session
4. Check Session Credits
5. Display Account Details
6. Exit
Choose an option: 4
--- Check Session Credits ---
Your current session credits are: 13
--- Fitness Membership Management System ---
1. Create Membership Account
2. Add Session Credits
3. Book a Training Session
4. Check Session Credits
5. Display Account Details
6. Exit
Choose an option: 5
--- Display Account Details ---
Member Name: Sandile Ndlovu
Membership Number: 12345678
Session Credits: 13
--- Fitness Membership Management System ---
1. Create Membership Account
2. Add Session Credits


Microsoft Visual Studio Debu
6. Exit
Choose an option: 4
--- Check Session Credits ---
Your current session credits are: 13
--- Fitness Membership Management System ---
1. Create Membership Account
2. Add Session Credits
3. Book a Training Session
4. Check Session Credits
5. Display Account Details
6. Exit
Choose an option: 5
--- Display Account Details ---
Member Name: Sandile Ndlovu
Membership Number: 12345678
Session Credits: 13
--- Fitness Membership Management System ---
1. Create Membership Account
2. Add Session Credits
3. Book a Training Session
4. Check Session Credits
5. Display Account Details
6. Exit
Choose an option: 6
Thank you for using the Fitness Membership Management System. Goodbye!
=== Code Execution Successful ===
C:\Users\Sandile\source\repos\Fitness Membership System\x64\Debug\Fitness Membership System.exe (process 14940) exited with code 0 (0x0).
Press any key to close this window . . .

Visual Studio 2022.zip

https://1drv.ms/u/c/496e50603b0f41d2/Ecq9B0T1IoFGm8XgM3xKWnEB19Cgp9ZQ0wogNpGQqpQK4w?e=YgDdw0